INCH–POUND

MIL–T–83483B
27 May 1994
SUPERSEDING
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23 May 1978

MILITARY SPECIFICATION

THREAD COMPOUND, ANTISEIZE, MOLYBDENUM DISULFIDE–PETROLATUM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements of one type of antiseize thread compound, composed of molybdenum disulfide–petrolatum (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

VV–P–236	Petrolatum, Technical
PPP–C–96	Cans, Metal, 28 Gage and Lighter
PPP–B–636	Boxes, Fiberboard

MILITARY

MIL–M–7866	Molybdenum Disulfide, Technical – Lubrication Grade

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: ASD/ENOS, Building 125, 2335 Seventh Street, Suite 6, Wright–Patterson AFB OH 45433–7809 by using the self–addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

AMSC/NA FSC 8030

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STANDARDS
FEDERAL

FED–STD–313	Materials Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities
FED–STD–791	Lubricants, Liquid Fuels and Related Products, Method of Testing

MILITARY

MIL–STD–129	Sampling Procedures and Tables for Inspection by Attributes
MIL–STD–290	Packaging, Packing, and Marking of Petroleum and Related Products

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111–5094, phone (215) 697–2667.)

2.2 Non–Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D217	Cone Penetration Of Lubricating Grease
ASTM D4057	Manual Sampling of Petroleum and Petroleum Products

(Application for copies should be addressed to the Society of Automotive Engineers Inc., 400 Commonwealth Drive, Warrendale PA 15096; phone (412) 776–4841.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other information services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS.

3.1 Material. The physical composition of the thread compound shall be as specified in table I.

TABLE I. PHYSICAL COMPOSITION

Ingredient	Percent by weight (minimum)	(maximum)
Molybdenum disulfide	48	52
Petrolatum	Balance	Balance

3.1.1 Petrolatum. The petrolatum shall conform to federal specification VV–P–236.

3.1.2 Molybdenum disulfide. The molybdenum disulfide shall conform to MIL–M–7866.

3.2 Functional characteristics.

3.2.1 Worked penetration. The worked penetration of the thread compound shall be 170 or above penetration range, but not to exceed 260 when tested as specified in 4.3.1.

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3.2.2 Stability. The thread compound shall show no separation when tested as specified in 4.4.4.

3.2.2 Toxicity. The thread compound shall have no adverse effect on the health of personnel when used for its intended purpose. The compound shall contain no elements which produce noxious vapors in such concentration as to irritate personnel during formulation or use under conditions of adequate ventilation when exercising caution to avoid prolonged contact with the skin and while observing Occupational Safety and Health Administration (OSHA) guidelines. Questions pertaining to the toxic effects shall be referred by the procuring activity to the appropriate departmental medical service who will act as an advisor to the procuring activity, Material safety data sheets shall be prepared and submitted in accordance with FED–STD–313.

3.3 Workmanship. The thread compound shall be free from cakes or lumps and hard, gritty particles. There shall be no separation of the mixture when tested as specified in section 4.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as quality conformance inspection (see 4.3).

4.3 Quality conformance inspection. Quality conformance inspection shall consist of sampling plans A and B.

4.3.1 Batch lot. A lot shall consist of 500 pounds or less of compound manufactured at one time from one batch of compound (see 3.1) offered for delivery at one time.

4.3.2 Sampling plan A. A one–pound sample of the compound (see 3.1) shall be selected at random from each inspection lot in accordance with ASTM D 4057, FED–STD–791 and subjected to the tests specified in 4.4.

4.3.3 Sampling plan B. A random sample of filled containers fully prepared for delivery shall be selected in accordance with table I. This samples shall be subject to the inspections specified in 4.4.2. If any sample fails, the lot shall be rejected.

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Table I. Sampling Plan

Lot Size	Sample Size
1–4	all
5–50	5
51–90	7
91–150	11
151–280	13
281–500	16
501–12,000	19
[acceptance number is zero, (c + 0)]	

4.4 METHODS OF INSPECTION.

4.4.1 Inspection. Inspection shall be in accordance with method 9601 of FED–STD–791.

4.4.2 Examination of filled containers. Each sample, selected in accordance with 4.3.3, shall be visually examined for appearance of the compound (see 3.3), for defects of construction of the container and the closure, for evidence of leakage, for unsatisfactory markings and all other preparation for delivery requirements of section 5. Each sample filled container shall also be weighed to determine the amount of contents. Any container in the sample having one or more defects, or under the required fill, shall be rejected.

4.4.3 Penetration. The normal worked penetration of the thread compound shall be determined in accordance with ASTM D217.

4.4.4 Stability. The stability of the thread compound shall be determined by placing 100 grams of the compound in each of the two cone–shaped centrifuge tubes, and centrifuging at 1500 rpm for 1/2 hour. The apparatus to be used for this test is described in FED–STD–791, Method 3003. Separation shall be defined as droplets or layer of oil appearing on the surface of the compound after centrifuging.

4.4.5 Rejection. Failure of any sample selected in accordance with 4.3.2 to pass any of the tests of 4.4 shall be cause for rejection of the lot represented.

5. PACKAGING.

5.1 Preservation – packaging. Preservation–packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Thread compound shall be packaged in cans conforming to PPP–C–96, Type V, Class 2, round.

5.1.1.1 Level C. Thread compound shall be packaged in accordance with the manufacturer's commercial practice.

5.1.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.1.2.1 Level A. Thread compound shall be packed 48 (one–pound cans) in boxes conforming to PPP–B–636, Type I, Class 2, Grade 3, 5, or 6.

5.1.2.2 Level B. Thread compound shall be packed 48 (one–pound cans) in boxes conforming to PPP–B–636, domestic type.

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5.1.2.3 Level C. Thread compound packaged as specified in 5.1, shall be packed in a manner to insure carrier's acceptance and safe delivery to destination. Containers shall be in accordance with rules and regulations of carriers applicable to the mode of transportation.

5.1.2.4 Marking precautionary. The following markings shall appear on each package and shipping container:

DANGER: THIS MATERIAL IS AN ELECTRICAL CONDUCTOR.
APPLY LIGHT COATING TO LOWER SPARK PLUG THREADS ONLY.
MATERIAL MUST NOT COME IN CONTACT WITH SPARK PLUG TERMINAL
OR ELECTRODES. DO NOT USE IN OXYGEN SYSTEMS – EXPLOSION MAY RESULT.

5.2 Marking unit and shipping containers. In addition to special marking required, unit and shipping containers shall be marked in accordance with MIL–STD–129.

6. NOTES.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This thread compound is particularly suitable for use on aircraft engine spark plugs and threaded fasteners and fittings at temperatures below 800°F. This compound contains molybdenum disulfide which at higher temperature conditions (1000°F and above) may induce hot corrosion of fastener or contiguous materials. Accordingly, its use at higher temperatures shall be avoided. MIL–T–5544 thread compound should be used in lieu of products directed by this specification in these high temperature applications for Air Force use. It should not be used in lieu of sealing compound, pipe joint and thread, lead free, general purpose conforming to TT–S–1732.

6.2 Acquisition requirements. Acquisition documents should specify the following information:

a. Title, number and date of this specification.
b. Applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Unit of purchase. The material should be purchased by weight, the unit being one pound.

6.4 Subject term (key word) listing.

Molybdenum
Thread compound
Petrolatum

6.5 Changes from previous issues. Marginal notations are not used in this revisions to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army – MR
Navy – SH
Air Force – 11

Preparing Activity:
Air Force – 11
(Project 8030–0686)

Review activities:

Air Force – 99

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